

Question	Answer	Mark	Mark scheme
13	Proof shown	C2	for complete argument, eg $n(n - 1)$ is the product of two consecutive integers and must be even as either n or $n - 1$ must be even or gives correct reasoning for n odd and n even n odd: odd $\times$ odd = odd and odd $-$ odd = even n even: even $\times$ even = even and even $-$ even = even or n odd: $(2n + 1)^2 - (2n + 1) = 4n^2 + 2n = 2(2n^2 + n)$ n even: $(2n)^2 - (2n) = 4n^2 - 2n = 2(2n^2 - n)$
		(C1	for factorising, eg $n(n - 1)$ OR gives correct reasoning for n odd or n even OR gives a partial explanation using n odd and n even, eg odd² $-$ odd = even and even² $-$ even = even)